Review Questions

1. Encircle the correct answer:

(i) A localized group of organisms that belong to the same species is called a:

(a) Biosphere (b) Community

(c) Ecosystem (d) Population

(ii) Increased quantity and quality of fish production:

(a) Fisheries (b) Farming

(c) Animal husbandry (d) Forestry

(iii) Study of remote past organic life, with the help of fossils.

(a) Entomology (b) Paleontology

(c) Taxonomy (d) Histology

(iv) Laws and techniques of physics are applied to explain the living processes of life.

(a) Biometry (b) Biostatistics

(c) Biophysics (d) Bio-economics

(v) Choose the incorrect statement:

(a) Six elements such as C, H, O, N, S and P are called basic elements of life.

(b) Foundation of life based on chemicals.

(c) Members of different species form population.

(d) Part of earth where life exists is called biosphere

(vi) Science of diagnosis and treatment of diseases.

(a) Agriculture (b) Medicine

(c) Surgery (d) Both B and C

(vii) Similar cells combine together to form:

a) Organs (b) System

(c) Tissue (d) Body

(viii) Scientific name of frog is:

(a) Palaeon (b) Rana tigrina

(c) Periplaneta (d) Pheretima

(ix) Select the correct sequence of biological organization

(a) Atom → Cell → Tissue → Molecule → Organ

(b) Atom → Tissue → Cell → Molecule → Organ

(c) Atom → Molecule → Cell → Tissue → Organ

(d) Atom → Cell → Molecule →Tissue → Organ

(x) *Volvox* is a polyphyletic genus of

(a) Green algae (b) Red algae

(c) Brown algae (d) None of these

2. Fill in the blanks

(i) Techniques for manipulation of gene to achieve desirable characters are called_____.

(ii) Distribution of different living organisms in different regions of the world_____.

(iii) Part of agriculture for the development of new varieties of plant, and their fruit is _____.

(iv) Bio elements considered as vital for life are _____ in members.

(v) Members of different species living in specific habitat are called _____.

(vi) The Muslim scientists who identified many diseases like tuberculosis, meningitis and other such inflammations was _____.

(vii) Part of earth where life exists is called_____.

- Conclusion is where all the results from the experiment are analyzed and a determination is reached about the hypothesis.
- Theories are the well-tested and highly reliable scientific explanations of natural phenomena.
- A scientific law is a uniform or constant fact of nature.
- Mathematical biology is a field of research that examines mathematical representations of biological systems.

Review Questions

1. Encircle the correct answer:

(i) Select the correct sequence of biological method.

(a) Law → Theory → Reasoning → Hypothesis

(b) Hypothesis → Theory → Law → Reasoning

(c) Hypothesis → Reasoning → Theory → Law

(d) Law → Hypothesis → Reasoning → Theory

(ii) Select the odd one

(a) Theory (b) Law

(c) Hypothesis (d) Ratio

(iii) Field of research that examines mathematical representations of biological system called.

(a) Ratio (b) Mathematical biology

(c) Proportion (d) Law

(iv) Biological method involves all of the following except.

(a) Data collection (b) Observation

(c) Experiment (d) Proportion

(v) Scientific reasoning moves from specific to general.

(a) Inductive (b) Deductive

(c) Observation (d) Both a and b

(vi) Quantitative observation is the use of

(a) Senses (b) Equipment

(c) Guess (d) Ratio

(vii) Equation stating that two ratios are equal

(a) Ratio (b) Proportion

(c) Guess (d) Senses

(viii) The comparison of two values is called

(a) Ratio (b) Proportion

(c) Graph (d) Table

(ix) What is a hypothesis?

(a) The same thing as an unproven theory.

(b) A tentative explanation that can be tested and is falsified.

(c) A verifiable observation.

(d) A fact based on quantitative data that is falsified.

(x) In data organization, which method is mostly useful?

(a) Table (b) Graph

(c) Ratio (d) Both a and b

2. Fill in the blanks:

(i) Problem solving approach of biology and other sciences called_____.

(ii) Solution of biological problem starts with_____.

(iii) The key component of the scientific process is _____.

(iv) Scientific reasoning based on "if-then" statement called _____.

Review Questions

1. Encircle the correct answer:

(i) Which of the following is the correct way of writing a scientific name of an organism?

(a) *Houbara bustard* (b) E.coli

(c) Alium Cepa (d) canis lupis

(ii) Select the mismatched

(a) Plantae → Pteridophyta (b) Fungi → Mucor

(c) Protista → Paramecium (d) Animalia → Amoeba

(iii) Identify the correct sequence of classifying an organism.

(a) Species → Genus → Kingdom → Phylum → Class → Order → Family

(b) Kingdom → Phylum → Class → Order → Family → Genus → Species

(c) Kingdom → Phylum → Class → Family → Order → Genus → Species

(d) Species → Genus → Class → Phylum → Order → Kingdom → Family

(iv) All are involve in classification except

(a) Analogous (b) Homologous (c) Cytology (d) Genetics

(v) In the taxonomic hierarchy choose the term which encompasses all other in the list

(I) Genus (II) Species (III) Order (IV) Class

(a) I and II (b) II (c) II and III (d) IV

(vi) In four kingdom classification, kingdom Metaphyta includes all of the following except.

(a) Algae (b) Angiosperm

(c) Gymnosperm (d) Bryophyta

(vii) In five kingdom system virus placed in

(a) Monera (b) Protista

(c) Plantae (d) None of these

(viii) Biological name of cat

(a) Felis catus (b) Azadirachta indica

(c) Alium Cepa (d) Canis lupis

(ix) Members of which kingdom have cell wall and are all heterotrophic

(a) Monera (b) Protista (c) Plantae (d) Fungi

(x) Biodiversity is effected with

(I) Pollution (II) Deforestation (III) Over hunting

(a) I only (b) II only (c) I and II (d) I, II and III

2. Fill in the blanks:

(i) Degree of variation within or among the species exists on different regions of the earth called _____.

(ii) _____ Species are given scientific names.

(iii) _____ Structures have different functions (physiology) but having similar internal architecture.

(iv) Science of classification is called _____.

(v) Most protists are _____.

(vi) _____ are achlorophyllous and absorptive living organism.

(vii) _____ is non-cellular obligate parasite.

(viii) Scientific name of any organism should be _____ when printed.

(ix) Cutting down of trees called _____.

(x) Animals which are at the risk of extinction in future called _____.

3. Define the following terms:

(i) Analogous (ii) Classification (iii) Species

(iv) Family (v) Metazoa (vi) Mycelium

(vii) Hyphae (viii) Genus (ix) Endangered species

(x) Kingdom

Review Questions

1. Encircle the correct answer:

(i) What is responsible for the high resolution of the electron microscope?

(a) High magnification

(b) Short wavelength of the electron beam

(c) Use of heavy metals strains

(d) Very thin section

(ii) What is a function of the rough endoplasmic reticulum?

(a) Aerobic respiration (b) Intracellular digestion

(c) Synthesis of steroids (d) Synthesis of protein

(iii) Which statement about the fluid mosaic model of membrane structure is correct?

(a) The less unsaturated the fatty acid, the more fluid nature.

(b) The more unsaturated the fatty acid, the more fluid nature.

(c) Higher the temperature, less fluid nature.

(d) The lower the temperature, more fluid nature

(iv) Which process allow movement in and out of cell

I. Osmosis

II. Diffusion

III. Active transport

(a) I only (b) I and II only

(c) II and III only (d) I, II and III

(v) All are postulates of cell theory except

(a) New cell is derived from pre-existing cells.

(b) Cell does not contain the hereditary material.

(c) All living organisms are made up of one or more cells.

(d) Cell is the fundamental unit of life

(vi) Secondary wall is made up of

(a) Pectin and cellulose (b) Cellulose and protein

(c) Cellulose and lignin (d) Lignin and pectin

(vii) Select the odd one

(a) Active transport (b) Diffusion

(c) Facilitated diffusion (d) Osmosis

(viii) Trace the correct pathway of protein produce from protein factories

(a) RER → Ribosome → Golgi body → Lysosome

(b) Ribosomes → RER → Golgi body → Lysosome

(c) Golgi body → RER → Ribosome → Lysosome

(d) RER → Ribosome → Lysosome → Golgi body

(ix) Cell organelle found in animal cell and help intracellular digestion

(a) Lysosome (b) Ribosomes

(c) Mitochondria (d) Golgi apparatus

(x) Select the mismatched

(a) Plastids → Storage of chemicals

(b) Centriole → Help in cell division

(c) Ribosomes → Synthesis of steroids

(d) Mitochondria → Synthesis of ATP

2. Fill in the blanks:

(i) Microscopes are instrument designed to produce _____ visual image.

(ii) Resolution of a microscope is defined as the smallest distance between_____ points.

(iii) Magnification of a light microscope is formed by using mixture of the power of the eyepiece and the _____ lens.

(iv) Electron has a much shorter wavelength than visible light, and this allows electron microscopes to produce _____ images.

(v) In plants, the cell wall is composed mainly of strong fibers of _____.

- Cell-death occur in two ways:
  (i) Apoptosis; programmed cell death, which help in proper control of development.
  (ii) Necrosis; cell death occur by external factor or accidental cell-death.

Review Questions

1. Encircle the correct answer:

(i) Which processes involve mitosis?
(a) Growth, reduction division and asexual reproduction.
(b) Growth, repair and asexual reproduction.
(c) Growth, repair and semiconservative replication.
(d) Growth, reduction division and repair

(ii) What occurs in metaphase of mitosis?
(a) Chromosome line up on the equator of the cell.
(b) Chromatids reach the poles of the spindle.
(c) Chromatids separate and move to opposite poles
(d) Chromosomes start to coil up and become visible.

(iii) Select the mismatched
(a) Anaphase → Chromatids move
(b) Prophase → Centriole move
(c) Telophase → Nuclear membrane disappear
(d) Metaphase → Chromosome line up.

(iv) Which process occurs during prophase of mitosis in an animal cell?
(a) Division of centromeres (b) Formation of chromosomes
(c) Replication of DNA (d) Separation of centrioles

(v) Sequence of physiological changes by which cell commit suicide.
(a) Apoptosis (b) Necrosis
(c) Autophagy (d) Both “a” and “b”

(vi) Choose the incorrect statement regarding meiosis.

(a) Maintain chromosome number from generation to generation.

(b) Reduces the diploid number of chromosomes to half

(c) Takes place in germ cell to produce gametes

(d) Production of new somatic cells from germ cells.

(vii) A type of cell division in which spore mother cell produce spores.

(a) Amitosis (b) Mitosis

(c) Meiosis (d) Both "a" and "c"

(viii) Stage of mitosis in which chromatids reach the poles and their movement ceases.

(a) Prophase (b) Metaphase

(c) Anaphase (d) Telophase

(ix) Stage of meiosis in which centromeres shorten and the paired chromatids are pulled away from one another.

(a) Anaphase-II (b) Metaphase-II

(c) Telophase-II (d) Prophase-II

(x) Phenomenon in which pair of homologous chromosome fails to separate called.

(a) Non-disjunction (b) Terminalization

(c) Synapsis (d) Linkage

2. Fill in the blanks:

(i) Chromosomes are thread like structures appear at the time of _____.

(ii) Sequence of changes which occurs between one cell division and the next is called_____.

(iii) The two chromatids are attached to each other at_____.

(iv) Chromosomes in the cell which are similar to each other in shape and size are called_____.

Review Questions

1. Encircle the correct answer:

(i) All are characters of enzymes except.
(a) Enzyme speed up a biochemical reaction.
(b) Enzymes are sensitive to minor change in pH
(c) Enzyme activity enhanced by inhibitors
(d) Enzyme portion where substrate attach called active site

(ii) Enzymes are
(a) Steroid in nature (b) Protein in nature
(c) Lipid in nature (d) Carbohydrate in nature

(iii) Metabolic reactions are
(I) Constructive reactions
(II) Destructive reactions
(III) Inhibiting reactions
(a) I only (b) I and II only (c) I, II and III (d) II and III only

(iv) The point where the enzyme is most active is known as
(a) Neutral pH (b) Acidic pH
(c) Basic pH (d) Optimum pH

(v) Active site continuously changes it shapes until the substrate do not bind to it, is statement of.
(a) Induce fit model (b) Lock and key model
(c) Fluid mosaic model (d) Both “a” and “b”

(vi) Select the mismatched
(a) Proteases → Carbohydrate (b) Lipases → Lipids
(c) Trypsin → Protein (d) All are correctly matched

(vii) Chemical reaction requires particular conditions to carry down at proper rate, especially.
(a) Temperature and Nature (b) Nature and Pressure
(c) Nature and Structure (d) Temperature and Pressure

(viii) All are factors affecting enzyme activity except.

(a) pH (b) Substrate concentration

(c) Organic solvent (d) Temperature

(ix) Rate of reaction will increase when temperature

(a) Increases (b) Decreases

(c) Below 10°C (d) Both "a" and "c"

(x) Choose the correct statement regarding lock and key model.

(a) Enzyme and substrate posses' specific complementary geometric shapes.

(b) Active site of enzyme is flexible

(c) Active site continuously changes

(d) All above statements are correct.

2. Fill in the blanks:

(i) There are _____types of metabolic reactions.

(ii) Enzymes catalyze chemical reaction by _____ the activation energy.

(iii) Presence of enzyme does not affect the nature or properties of _____.

(iv) In constructive reaction _____ molecules are formed.

(v) Activity of enzymes can be enhanced by _____.

(vi) Small portion of enzyme where substrate attach with enzyme called_____.

(vii) Enzyme activity decreased by _____.

(viii) As temperature increases, initially the rate of reaction will_____.

(ix) Extreme changes in pH can cause enzymes to _____.

(x) In the human body there are more than _____ known enzymes.

3. Define the following terms:

(i) Substrate (ii) Active site (iii) Inhibitor

(iv) Activator (v) Anabolism (vi) Catabolism

(vii) Activation energy (viii) Cofactor (ix) Prosthetic group

(x) Coenzymes

Review Questions

1. Encircle the correct answer:

(i) In an oxidation process 14135KJ energy is release, how many moles of glucose consume during this process.

(a) 1 (b) 3 (c) 5 (d) 10

(ii) Stage of aerobic respiration takes place at the cristae of mitochondria called.

(a) Electron transport chain (b) Glycolysis

(c) Kreb's cycle (d) C3 cycle

(iii) In a process of cellular respiration 180 ATP molecules are produced, how many moles of glucose consume during this process.

(a) 2 (b) 5 (c) 8 (d) 10

(iv) Loss of electron and proton is called

(I) Oxidation reaction (II) Reduction reaction

(III) Redox reaction

(a) I only (b) I and II (c) II and III (d) I, II and III

(v) Each mole of ATP store energy

(a) 7.3 kcal/mole (b) 7.3kj/mole

(c) 17.3kcal/mole (d) 17.3kj/mole

(vi) Fundamental molecule produced during photosynthesis is

(a) Glucose (b) Amino acid (c) Fatty acid (d) Nucleotide

(vii) Light dependent reaction takes place in

(a) Stroma (b) Thylakoid (c) Cristae (d) Cisternae

(viii) Reaction in which solar energy is transferred to glucose from ATP and $NADPH_2$, takes place in stroma called

(I) Light reaction (II) Dark reaction

(III) Light dependent reaction

(a) I only (b) II only (c) I and II (d) II and III

(ix) Splitting of water in presence of light called

(a) Hydrolysis (b) Glycolysis

(c) Photolysis (d) None of these

(x) Splitting of glucose (glycolysis) release small amount of energy which is enough to generate

(a) 2ATP (b) 5 ATP (c) 18 ATP (d) 36ATP

2. Fill in the blanks:

(i) The only source of energy on earth is_____.

(ii) Conversion of free energy into different forms by living organisms is called_____.

(iii) In living organisms energy is stored in a special molecule called_____.

(iv) Plant utilizes simple inorganic molecules (CO_2 and H_2O) to prepare _____.

(v) Feeding sequences and relationships are called _____.

(vi) Photosynthesis is the only process which produces free O_2 by splitting_____.

(vii) Chloroplast is double membrane bounded organelle have semifluid protein containing membrane called_____.

(viii) In chloroplast different pigments absorb light of different _____.

(ix) The breakdown of food molecules to release energy is called _____.

(x) Each mole of glucose produce maximum energy i.e. _____.

3. Define the following terms:

(i) Bioenergetics (ii) Energy (iii) Oxidation reaction

(iv) Food chain (v) Granum (vi) Photolysis

(vii) Fermentation (viii) Stroma (ix) Aerobic respiration

(x) Pyruvic acid

- So many environmental hazard are also related to chemical fertilizers
- There are seven major classes of nutrients, Carbohydrates, proteins, fats, minerals, fibers, vitamins and water.
- Balanced diet is related to age, six and activity of human.
- Problems related to nutrition are grouped as malnutrition.
- Kwashiorkor is a severe difficulty in dietary protein.
- Marasmus characterized by energy deficiency sum of mineral difficulty diseases are Goiter, anemia.
- Effects of malnutrition are starvation, heart diseases, constipation, obesity.
- Digestion is the break down of complex food into simple absorbable nutrients.

Review Questions

1. Encircle the correct answer:

(i) Select the mismatched

(a) Protein → Amino acid (b) Carbohydrate → Glucose

(c) Fats → Starch (d) Nucleic acid → Nucleotide

(ii) Deficiency of vitamin-k cause

(a) Rickets (b) Anemia

(c) Scurvy (d) Beriberi

(iii) Period of rapid growth and development called

(a) Childhood (b) Adult hood

(c) Adolescence (d) Both “a” and “b”

(iv) Deficiency of dietary protein, causes an imbalance in the gastro-intestinal system

(a) Marasmus (b) Edema

(c) Diarrhea (d) Kwashiorkor

(v) An animal that migrates great distances would obtain the greatest benefit from storing its energy as

(a) Fats (b) Carbohydrates

(c) Protein (d) Minerals.

(vi) Which of the following vitamins is correctly associated with its use?

(a) Vitamin K → Production of white blood cells

(b) Vitamin C → Curing rickets

(c) Vitamin E → Protection of skin from cancer

(d) Vitamin A → Incorporated into the visual pigment of the eye

(vii) Which of the following statements describes pepsin?

(a) It is manufactured by the pancreas.

b) It helps stabilize fat-water emulsions.

(c) It splits maltose into monosaccharides.

(d) It begins the hydrolysis of proteins in the stomach.

(viii) Which of the following is true of bile salts?

(a) They are enzymes

(b) They are manufactured by the pancreas

(c) They emulsify fats in the duodenum

(d) They increase the efficiency of pepsin action.

(ix) In human digestive system trachea and oesophagus both connect to the

(a) Large intestine (b) Stomach

(c) Pharynx (d) Rectum

(x) All are sources of calcium except

(a) Red meat (b) Green leafy vegetables

(c) Broccoli (d) Nuts

Review Questions

1. Encircle the correct answer:

(i) The movement of organic materials (food) take place through

(a) Xylem (b) Vessels

(c) Tracheid (d) Phloem

(ii) What regulates the flow of water through the xylem?

(a) Passive transport by the endodermis

(b) The number of companion cells in the phloem

(c) The evaporation of water from the leaves

(d) Active transport by sieve-tube membrane

(iii) Phloem transport of sucrose can be described as going from "source to sink" which of the following would not normally function as a sink?

(a) Mature leaf (b) Storage organ

(c) Growing root (d) Both 'b' and 'c'

(iv) Human plasma proteins include which of the following?

(I) Fibrinogen (II) Haemoglobin (III) Albumin

(a) I only (b) II only (c) I and II (d) I and III

(v) Which of these are involved in the clotting of blood?

(a) Platelets (b) Haemoglobin

(c) Albumin (d) Globulin

(vi) In human heart blood returning from lungs first drains into the

(a) Left atrium (b) Left ventricle

(c) Right atrium (d) Right ventricle

(vii) Root hairs are most important to a plant because they

(a) Store starch (b) Contain xylem tissues

(c) Provide a habitat for nitrogen fixing bacteria

(d) Increase the surface area for absorption.

(viii) A type of circulatory system blood flow through the spaces among tissues called.

(I) Open type circulatory system

(II) Close type circulatory system

(III) Pulmonary circulatory system

(a) I only (b) II only

(c) I and II (d) II and III

(ix) Part of root between epidermis and endodermis called

(a) Xylem (b) Root hair

(c) Phloem (d) Cortex

(x) Movement of water molecules from the area of higher water potential to the area of lower water potential called

(a) Diffusion (b) Osmosis

(c) Active transport (d) Water potential

2. Fill in the blanks:

(i) Loss of internal water of plant in the form of vapours from aerial parts of plants is called_____.

(ii) In flowering plants xylem is made up of _____ types of tissues.

(iii) Root hair is long, thin and tubular structure, it increases surface area which increases the _____.

(iv) Opening and closing of stomata is also regulated by_____.

(v) A type of cancer that affects the blood, bone marrow and lymphatic system called_____.

(vi) Pumping of blood by the rhythmic pulsation of heart throughout the body is called_____.

(vii) Relaxation of heart muscles and allowing the chambers to fill with blood called_____.

(viii) Dead cells of xylem vessels arranged vertically have empty space inside called_____.